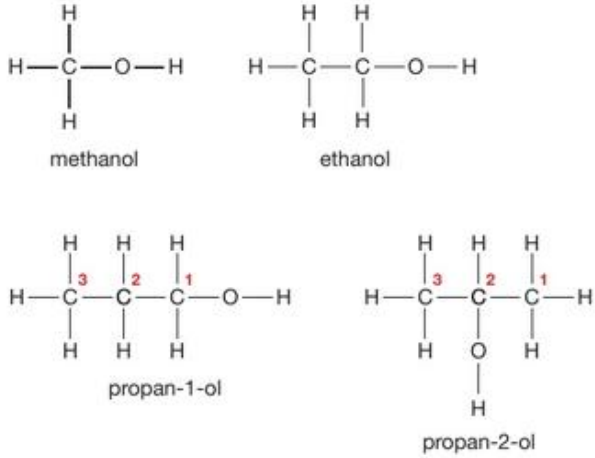
HINT

Ethanol is what is commonly known just as 'alcohol'.

Figure 22.21 shows four small alcohols.



▲ Figure 22.21 Four small alcohols

The names all end with '-ol', showing there is an –OH functional group in the molecule. Notice the way the number of carbons in the chain is counted exactly as before with 'meth', 'eth' and 'prop'.

There are two different forms of propanol. The –OH group can be attached to the end carbon of the chain (in propan-1-ol) or to the second carbon in the chain (as in propan-2-ol). These are *structural isomers* because there is no way that you can bend or twist one of the molecules to make the other. Bonds would have to be broken to get from one to the other.

EXTENSION WORK

It is possible to find a structural isomer of propanol that doesn't have an –OH functional group, for example $CH_3OCH_2CH_3$, which has the oxygen atom bonded between two carbon atoms rather than joined to a hydrogen atom. This compound is named methoxyethane and belongs to a homologous series of compounds called *ethers*.

END OF CHEMISTRY ONLY

SOME CHEMICAL REACTIONS OF ORGANIC COMPOUNDS

In this section we will look at how to classify the reactions of organic compounds. These reactions will be explained much more in the next few chapters.

COMBUSTION



▲ Figure 22.22 The burning of a camping stove

HINT

Take care when balancing this equation – do not forget the O in the alcohol!

All organic compounds that you will meet at International GCSE undergo combustion reactions. Combustion is just another way of saying *burning* and involves a reaction with oxygen.

Combustion of hydrocarbons in *excess oxygen* gives rise to *carbon dioxide* and *water*, together with the release of a large amount of heat energy. For example:

propane burning $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$

butene burning $C_4H_8(g) + 6O_2(g) \rightarrow 4CO_2(g) + 4H_2O(l)$

Propane is frequently used in camping stoves outdoors.

Alcohols undergo similar reactions:

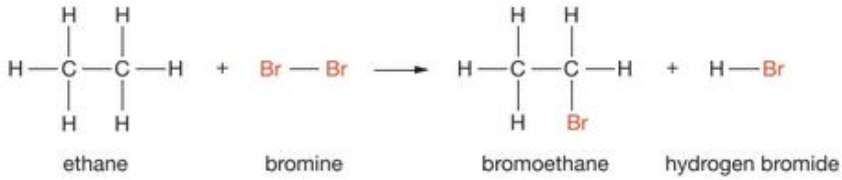
ethanol burning $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$

SUBSTITUTION

A **substitution** reaction occurs when an atom or group of atoms is *replaced* by a different atom or group of atoms. For example, the hydrogen atoms in an alkane can be replaced by halogen atoms. We will go into more detail on this reaction in Chapter 24.

ethane with bromine gas

$$CH_3CH_3(g) + Br_2(g) \rightarrow CH_3CH_2Br(g) + HBr(g)$$



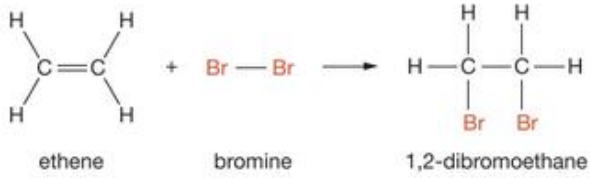
▲ Figure 22.23 A substitution reaction between an alkane and a halogen

ADDITION

In an **addition** reaction something is added to a molecule without taking anything away. Alkenes undergo addition reactions. For example, ethene reacts with bromine:

ethene with Br_2

$$CH_2CH_2(g) + Br_2(l) \rightarrow CH_2BrCH_2Br(l)$$



▲ Figure 22.24 An addition reaction between an alkene and a halogen.

CHAPTER QUESTIONS

Note: the questions below deliberately include some examples which are slightly beyond the level which you would expect to meet in an International GCSE exam.

SKILLS CRITICAL THINKING

6 7

1 a Write down the names of the following hydrocarbons:

- i CH_4
- ii $CH_3CH_2CH_3$
- iii $CH_3CH_2CH_2CH_2CH_2CH_3$
- iv $CH_3CH{=}CH_2$
- v $CH_2{=}CH_2$
- vi $CH_2{=}CHCH_2CH_3$
- vii $CH_3CH_2CH_2OH$
- viii $CH_3CH_2CHOHCH_3$

SKILLS INTERPRETATION

b Draw structural and displayed formulae for:

- i ethane
- ii propan-1-ol
- iii but-2-ene
- iv hexane
- v methanol
- vi 2-methylpropane
- vii methylpropene
- viii pent-1-ene

SKILLS CRITICAL THINKING

SKILLS INTERPRETATION

6 8

2 a Explain the term *structural isomerism*.

b There are two structural isomers of C_4H_{10}. Draw their structures and name them.

c There are five structural isomers of C_6H_{14}. Draw their structures and name them.

SKILLS CRITICAL THINKING

d How many structural isomers containing a carbon–carbon double bond can you find with a molecular formula C_4H_8? Name the straight chain isomers.

SKILLS INTERPRETATION

e Draw two isomers of C_4H_8 that do not contain carbon–carbon double bonds.

9

3 a Draw the structures for the following structural isomers of C_4H_9OH. (Note: this is beyond what you will be asked to do at International GCSE. The answer to the first one is shown to give you hints as to how to work out the others.)

i butan-1-ol (Answer: $CH_3CH_2CH_2CH_2OH$)

ii butan-2-ol

iii 2-methylpropan-1-ol

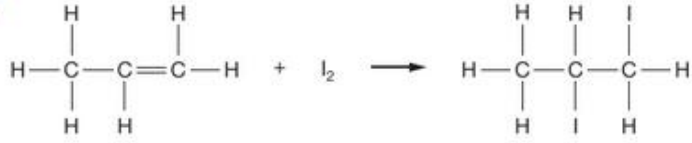
iv 2-methylpropan-2-ol

SKILLS CRITICAL THINKING

b Can you find two more structural isomers with the same molecular formula ($C_4H_{10}O$) which don't contain an –OH group?

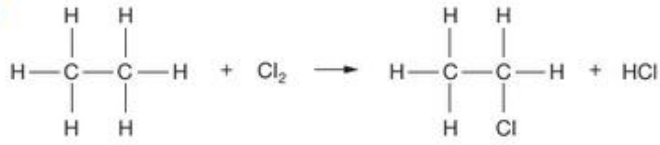
7

4 Classify the following reactions as substitution, addition or combustion.

a 

b $2C_8H_{18}(l) + 25O_2(g) \rightarrow 16CO_2(g) + 18H_2O(l)$

c $C_4H_8(g) + H_2(g) \rightarrow C_4H_{10}(g)$

d 

e $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$

f $CH_3CH_2CH_2CH_3(g) + Br_2(g) \rightarrow CH_3CH_2CHBrCH_3(g) + HBr(g)$

SKILLS CRITICAL THINKING 6

5 The table below shows the formulae of some organic compounds.

Compound	Formula
A	C_2H_6
B	$CH_3CH{=}CH_2$
C	$CH_3CH_2CH_2OH$
D	$CH_3CH_2CH_2CH_2CH_3$

a Select one compound from the table which is not a hydrocarbon. Explain your choice.

SKILLS INTERPRETATION 8

b Draw a dot-and-cross diagram to show the bonding in compound **A**.

c Draw the full displayed formula of compound **B**. Include all bonds in your drawing.

SKILLS CRITICAL THINKING 6

d Compounds **A** and **D** are from the same homologous series. State the general formula for this homologous series.

7

e Other than having the same general formula, state *one* other characteristic of members of the same homologous series.

SKILLS PROBLEM SOLVING 9

f Write a balanced chemical equation for the complete combustion of compound **D**.